

ChinookAnalytics: Final Analytical Report (Marts)

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Abstract

This report answers the analytical objectives defined in *Analysis Objectives*, using only the curated `marts` layer. All numbers below are derived from `mart` outputs produced by the SQL pipeline and validated against the `core` schema via reconciliation checks.

1 Executive Summary

- **Total revenue:** 2328.60
- **Total invoices:** 412
- **Customers:** 59 **Billing countries:** 24
- **Concentration snapshot:**
 - Top-1 artist share: 5.95%
 - Top-5 artists share: 21.57%
 - Top-10 artists share: 30.98%
 - Top-1 country share: 22.46%
 - Top-5 countries share: 58.78%
 - Top-10 countries share: 76.05%
 - Top-10 customers share: 19.38%
 - Top-20 customers share: 36.61%

High-level verdict: Revenue is stable over time with moderate concentration risk driven primarily by geography (country mix), while customer revenue is relatively diversified. Catalog concentration exists at the genre level (Rock dominates), but artist-level dependency is moderate.

2 Objective 1: Revenue Health Over Time

Questions answered: revenue evolution monthly/yearly, growth behavior, invoice value trend, volatility signals.

Key Results

- Revenue is reported at month grain (60 months).
- Total revenue equals 2328.60 across all months (validated).
- Yearly totals are relatively stable (no persistent multi-year growth trend).

Required Output Tables

- `marts.mart_revenue_time`: `month_date`, `invoices`, `revenue`, `avg_invoice_value`, `MoM`, `YoY`.
- Yearly roll-up computed from the monthly mart.

Actionable Interpretation (from marts)

- Use MoM and YoY to isolate months with unusually high acceleration or slowdown.
- Track volatility by computing the standard deviation of monthly revenue per year from `mart_revenue_time`.

3 Objective 2: Market Performance and Geographic Strategy

Questions answered: which markets drive revenue, market value per customer, and geographic concentration.

Top Markets by Revenue (from marts)

- USA: 523.06 (Top-1 country share = 22.46%)
- Canada: 303.96
- France: 195.10
- Brazil: 190.10
- Germany: 156.48

Market Concentration (from marts)

- Top-5 countries represent 58.78% of total revenue.
- Top-10 countries represent 76.05% of total revenue.

Interpretation

- Revenue is moderately concentrated geographically: a small set of markets drives most revenue.
- `marts.mart_market_performance` provides revenue per customer by country, enabling prioritization beyond pure volume.

4 Objective 3: Customer Economics and Revenue Concentration

Questions answered: customer LTV ranking, distribution shape, and Top-N dependency.

Customer Base Summary (from marts)

- Customers: 59
- Invoices per customer: min 6, max 7, average 6.98
- LTV range: 36.64 to 49.62
- Average LTV: 39.47 Median LTV: 37.62

Revenue Concentration (from marts)

- Top-10 customers represent 19.38% of total revenue.
- Top-20 customers represent 36.61% of total revenue.

Interpretation

- Customer revenue is not extremely concentrated: Top-20 accounts for 36.6%, not a majority.
- The narrow purchase frequency range (6–7 invoices/customer) indicates limited separation between “high-frequency” and “low-frequency” customers in this dataset.

5 Objective 4: Catalog Strategy (Artists and Genres)

Questions answered: top artists/genres by revenue and units, and concentration risk in the catalog.

Top Artists by Revenue (from marts)

- Iron Maiden: 138.60 (140 units)
- U2: 105.93 (107 units)
- Metallica: 90.09 (91 units)
- Led Zeppelin: 86.13 (87 units)
- Lost: 81.59 (41 units)

Top Genres by Revenue (from marts)

- Rock: 826.65 (835 units)
- Latin: 382.14 (386 units)
- Metal: 261.36 (264 units)
- Alternative & Punk: 241.56 (244 units)
- TV Shows: 93.53 (47 units)

Catalog Concentration (from marts)

- Top-1 artist share: 5.95%
- Top-5 artists share: 21.57%
- Top-10 artists share: 30.98%

Interpretation

- Artist dependency is moderate (Top-10 artists = 31%), indicating resilience against single-artist risk.
- Genre mix is more skewed: Rock is a dominant revenue block, indicating stronger genre dependence than artist dependence.

6 Objective 5: Invoice Structure and Revenue Composition

Questions answered: what defines high-value invoices; whether value is driven by diversity/lines.

Invoice Total Distribution (from marts)

	P25	Median	P75	P90	P95
Invoice Total	1.98	3.96	8.91	13.86	13.86

Top-10 Highest Invoices (from marts)

Invoice	Customer	Date	Country	Total
404	6	2025-11-13	Czech Republic	25.86
299	26	2024-08-05	USA	23.86
96	45	2022-02-18	Hungary	21.86
194	46	2023-04-28	Ireland	21.86
201	25	2023-05-29	USA	18.86
89	7	2022-01-18	Austria	18.86
88	57	2022-01-13	Chile	17.91
306	5	2024-09-05	Czech Republic	16.86
313	43	2024-10-06	France	16.86
208	4	2023-06-29	Norway	15.86

Interpretation

- High-value invoices are far above the median (Median = 3.96 vs Max = 25.86).
- `marts.mart_revenue_composition` provides decomposition fields (lines, distinct artists/genres/media types) to test whether outliers are driven by size vs diversity.

7 Objective 6: Dependency and Concentration Snapshot

Questions answered: dependency on top artists/countries/customers.

Snapshot Results (from marts)

Snapshot date: 2026-02-13

- Total revenue: 2328.60
- Artists: Top-1 = 5.95%, Top-5 = 21.57%, Top-10 = 30.98%
- Countries: Top-1 = 22.46%, Top-5 = 58.78%, Top-10 = 76.05%
- Customers: Top-10 = 19.38%, Top-20 = 36.61%

Interpretation

- Country dependency is stronger than customer dependency (Top-10 countries = 76%).
- Artist dependency is moderate (Top-10 artists = 31%).
- This indicates resilience against single-artist dependency, but meaningful exposure to market concentration.

8 Business Summary, Risks, and Action Plan

This section translates mart findings into business-oriented conclusions and a concrete plan of action.

8.1 Overall Business Summary

- **Revenue profile:** Stable across the observed period with no strong structural growth trend.
- **Market exposure:** Revenue is materially dependent on a limited set of countries (Top-10 = 76.05%).
- **Customer exposure:** Customer revenue is relatively diversified (Top-20 = 36.61%).
- **Catalog exposure:** Genre dependence is strong (Rock dominates), while artist dependence is moderate.
- **Basket behavior:** Invoice totals show heavy right-tail behavior (max 25.86 vs median 3.96), indicating occasional large baskets.

8.2 Key Risks and Points of Attention

1. Geographic concentration risk

A limited set of markets drives most revenue. This creates sensitivity to country-specific shocks (payment issues, regulation, currency, local competition).

2. Genre dependence

Rock is a dominant revenue block, suggesting the product mix may be less resilient to preference shifts.

3. Limited behavioral segmentation

With invoices per customer tightly clustered (6–7), it is difficult to detect meaningful loyalty tiers, churn risk, or frequency-based segmentation from this dataset alone.

8.3 Recommended Improvements to the Data Product

The current analytical layer is structurally correct and validated, but business decision power would increase with:

1. Pricing and promotion fields

Add discount, campaign, and net revenue fields at line level to quantify promotion elasticity.

2. Customer lifecycle fields

Add signup date, acquisition channel, and churn definition to enable retention analysis and cohort modeling.

3. Basket enrichment

Introduce variable quantities or bundle representation (quantity > 1, product bundles) to model basket scaling.

4. Currency normalization

If multi-country revenue is compared operationally, add currency and FX-normalized revenue for consistent market benchmarking.

8.4 Action Plan (Operational Next Steps)

A practical execution plan derived from the marts:

1. Market strategy (immediate)

Use `mart_market_performance` to identify the top revenue markets and compare revenue-per-customer. Prioritize 2–3 countries for focused growth programs while monitoring concentration.

2. Catalog strategy (near-term)

Use `mart_portfolio_genre` and `mart_portfolio_artist` to diversify the mix: promote underrepresented but high-value genres and reduce over-reliance on the dominant genre.

3. Outlier-driven investigation (near-term)

Use `mart_revenue_composition` to explain high-value invoices and determine whether they reflect diversity-driven baskets or single-category spikes.

4. **Monitoring layer (ongoing)**

Schedule regular refresh and track changes in: MoM/YoY growth, top country share, top artist share, top customer share.

9 Conclusion

The marts layer provides a consistent and validated business view of revenue across time, markets, customers, and catalog. The system confirms stable revenue behavior and highlights moderate concentration risk, primarily driven by geographic exposure.

The analytical deliverable is operationally complete: it defines objectives, materializes marts, and validates reconciliation. The recommended next step is to expand source-level business attributes (promotion, lifecycle, basket behavior) to increase decision leverage beyond descriptive analytics.